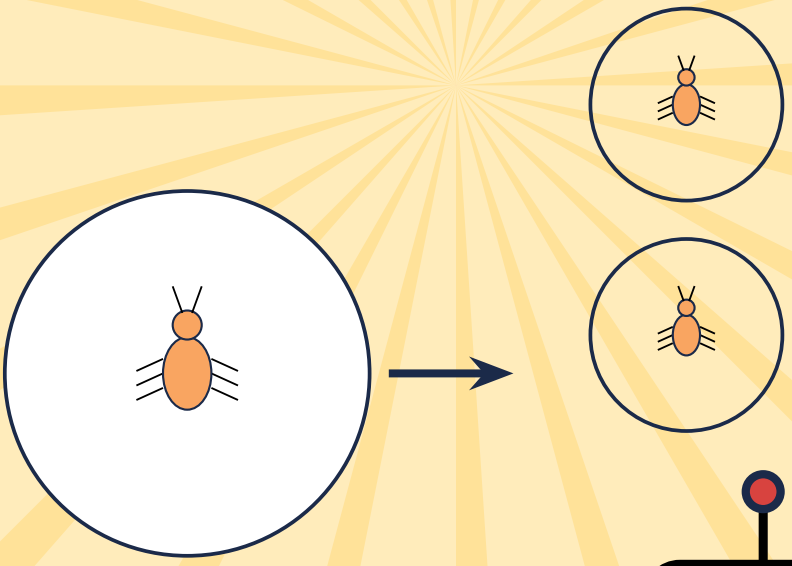
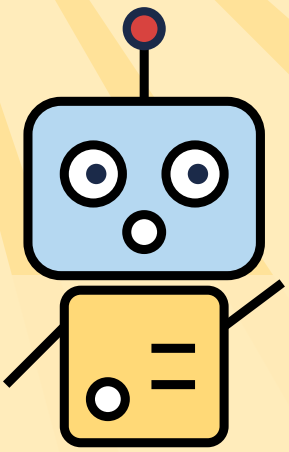
SCIENCE
Book 7

Sorting the Living



Look closely. Ask yes or no. Test your key.

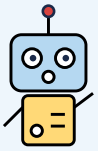
Puzzles, projects and Multix the guide robot



Ages 9 to 11

This book belongs to _____

Hello. I am Multix.

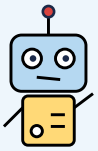


I am a guide robot. I read every instruction very carefully, which is why I sometimes get things wrong. When I make a mistake, see whether you can catch it. I never give away a puzzle answer.

How this book works

Every book in this series asks the same few questions, so once you have read this page you will know your way around.

- **Try it** boxes tell you what to do with a ruler, a cup, some string or a clock. You need nothing from a shop. Everything is at home or at school.
- **What did we keep?** After you simplify something, such as keeping the same conditions in a test or averaging several throws, you say what stayed and what was set aside. Simplifying is fine. Forgetting what you set aside is not.
- **What makes this correction trustworthy?** When you fix or improve an answer, a box says what you checked, what the check gives you reason to trust, and what it does *not* show.
- **Circles** show how a number was made. Page 4 explains how to read them, and Book 1 introduced them.
- **Puzzles** come at the end of each chapter. Try them before you look at the answers on page 45.
- **Slop check** is a page near the end. Slop is hasty, gaudy work that forgot to take care. You will learn to spot it and to fix it.



Here is a fact about me. I sorted a pile of buttons and announced the one right way to sort them. Chapter 1 will explain why I should be embarrassed.

In this book

1	Many Ways to Sort	5
2	Features That Matter	10
3	A Yes-or-No Key	16
4	Trading Keys	21
5	Groups Inside Groups	27
6	When Something Does Not Fit	32
⋆	Slop check	38
⋆	Answers	45
⋆	Words, and notes for grown-ups	47

Try it: Gather your kit

You will use these things across the six chapters. Everything is found at home or at school. Measuring comes from Book 1 and fair testing from Book 2, and both come back here.

- a ruler marked in mm
- twelve buttons, alike or not
- some leaves or shells
- a pencil and notebook
- paper and scissors
- a friend to trade keys with
- a camera, or paper to draw on
- some small toys or coins

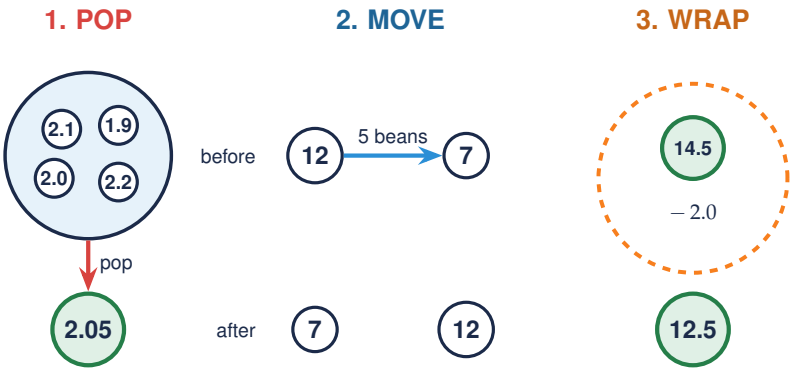
■ **Adult help:** Never touch or pick a plant you cannot name, and wash your hands after handling leaves. Watch animals from a distance and never pick them up. Keep buttons and small things away from little children and pets. Nothing here needs a flame or mains electricity.

Units in this book

We use the metric system: centimetres (cm), metres (m) and seconds (s). For sorting we mostly count things, and we measure lengths in centimetres when a feature needs it. There are 10 mm in 1 cm and 100 cm in 1 m. If you are used to inches, one inch is about 2.5 cm. We will use that only in side notes.

Reading the circles

A circle holds a group of things together. There are three things you can do with circles in this series.

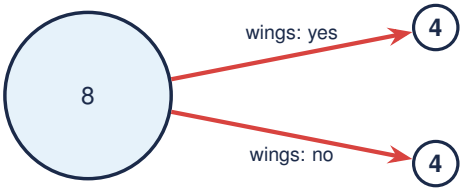


Pop Squash a group down to one value that sums it up, such as an average. A pop is a *summary*, so it always loses something. Every time we pop, we say what the summary keeps and what it loses.

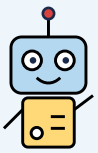
Move An arrow that carries something from one circle to another, such as beans or water. Nothing is summarized. You will see more of moves in later books.

Wrap When you correct something, draw a new dashed circle around the old one and show what changed. The old work stays visible, so anyone can check it.

A group is a circle that splits in two



One yes-or-no question splits one circle into two. A group inside a group is a smaller circle inside a bigger one.



A pop is not a lie, but it is a shortcut. Any time I tell you a pop, I will tell you what it threw away.

1 Many Ways to Sort

Choosing the feature that makes the groups

Dump a box of buttons on a table and you will want to sort them. But sort by what? You could sort by colour, by size, by how many holes they have, or by whether they are shiny. Each choice gives different groups from the very same buttons.

A feature you choose to sort by is a simplification. It keeps one thing about each button and sets aside everything else.

Try it: Sort twelve buttons two ways

You need: twelve buttons or small things such as coins, shells or toy bricks, and a sheet of paper.

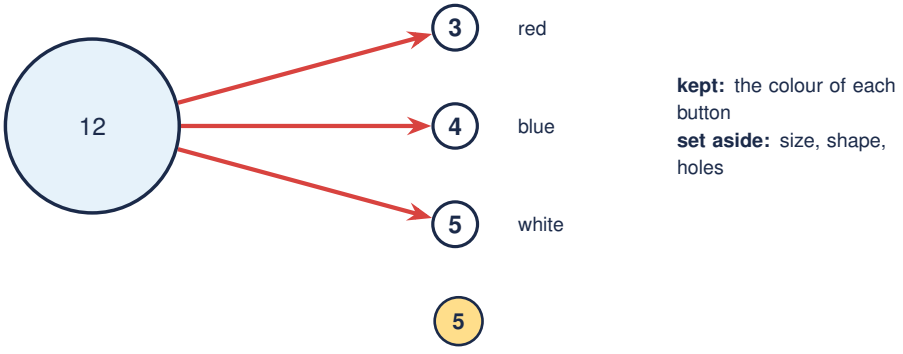
1. Sort the twelve things into groups using **one** feature, such as colour. Count each group.
2. Mix them up and sort again using a **different** feature, such as number of holes. Count each group.
3. Check that each sort adds up to twelve.

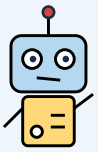
Feature	Group 1	Group 2	Group 3	Group 4	Total
First sort:					
Second sort:					

■ **Adult help:** Buttons and other small things can be swallowed by little children and pets. Work at a table and put them away afterwards.

Practice numbers, not from a real pile

By colour: red 3, blue 4, white 5. Total $3 + 4 + 5 = 12$.
By holes: two holes 7, four holes 5. Total $7 + 5 = 12$.
Both sorts used all twelve buttons exactly once. A button that is red *and* has two holes sits in one group in each sort, but never in two groups of the same sort.





There is one correct way to sort a pile of buttons, and I know it: by colour. Anything else is wrong. I sorted them first. Right?

What makes this correction trustworthy?

What we checked. We counted each group and added the groups to be sure every button was used once.

What this check gives us reason to trust. Each sort added to twelve, so no button was missed or counted twice.

What it does not show. That one way of sorting is the right one. Which feature is best depends on what you want to find out.

What did we keep?

We turned a pile of buttons into groups. **We kept** _____ **and we set aside** _____.

Puzzle 1.1: Count check

Buttons sorted by colour: red 3, blue 4, white 5. Total _____. Does it match twelve? _____

Puzzle 1.2: A missing button

A student sorted 12 buttons by holes into groups of 7 and 4. Total _____. What went wrong? _____

Puzzle 1.3: How many groups?

By colour the buttons made 3 groups. By holes they made 2. Which sort made more groups? _____

Puzzle 1.4: Choose features

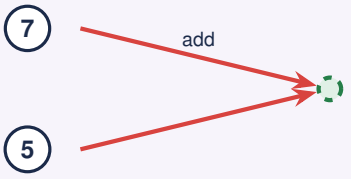
List three features you could use to sort a pile of shoes. _____

Puzzle 1.5: One right way?

Multix says there is one correct way to sort buttons. Write why that is not true. _____

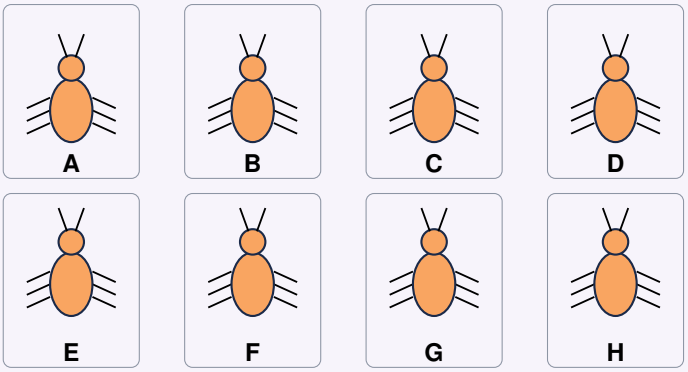
Puzzle 1.6: Draw the circles

Twelve buttons are sorted by holes into 7 and 5. Fill in the empty circle for the total.



Puzzle 1.7: The bug tray

Here is a tray of eight made-up bugs. Sort them by wings. How many have wings? _____ How many do not? _____



Puzzle 1.8: A group of one

Can a group have only one thing in it? _____ Give an example. _____

Puzzle 1.9: Both groups?

A red button has four holes. Can it be in both the "red" group and the "four holes" group? Explain. _____

Puzzle 1.10: Your sort

Sort ten things at home. Name the feature and the groups, with their counts.

_____

Sort log

Record two different sorts of the same pile. Fill in what you used and what you found.

Sort	Feature	Groups and counts	Total
First			
Second			
Third			

Which sort was easiest to do? _____

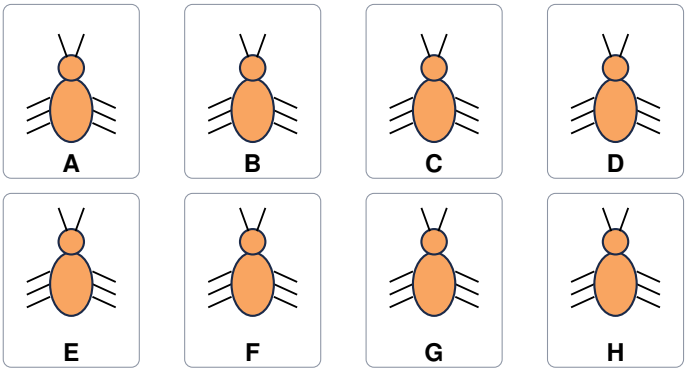
Which sort told you the most about the pile? _____

Draw the groups of your favourite sort as circles.

2 Features That Matter

Picking a feature that really separates things

Some features split a pile in a useful way and some do not. Here is the tray of eight made-up bugs again.



Look at the bugs. A feature is a **good splitter** when some bugs have it and some do not, and when everyone can tell which is which.

Feature	How many bugs have it	Good splitter?
Has wings	4 of 8	yes, a clear 4 and 4
Has spots	4 of 8	yes
Has long antennae	4 of 8	yes
Has six legs	8 of 8	no, it splits nothing
Is “nice looking”	depends on who looks	no, people disagree

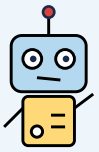
Try it: Rate the features

For each feature, count how many of the eight bugs have it. Then say whether it is a good splitter.

Feature	Bugs that have it	Good splitter? (yes/no)
Wings		
Spots		
Long antennae		
Six legs		

Check with real counts

Wings: A, B, E, F have wings, so 4 of 8. Spots: A, C, E, G, so 4 of 8. Long antennae: A, B, G, H, so 4 of 8. Six legs: all 8.
A feature that every bug has, or no bug has, cannot split the pile. It keeps nothing about the difference between bugs.



The best feature is the one most bugs have, because then I can sort most of them at once. Six legs has all eight bugs, so it is the best of all. Is it?

What makes this correction trustworthy?

What we checked. We counted each feature on every bug and checked that people could tell who has it.

What this check gives us reason to trust. Wings, spots and long antennae each gave a clear split of 4 and 4.

What it does not show. That these are the *best* features for every purpose. A feature matters only for a particular question.

What did we keep?

We chose wings to sort the bugs. **We kept** _____ **and we set aside** _____.

A clear feature, or a fuzzy one

"Is it big?" is fuzzy, because people will disagree about a middle-sized bug. "Is it longer than 2 cm?" is clear, because anyone with a ruler gets the same answer. Book 1 showed how a ruler and a rule turn a vague idea into a clear one.

Try it: Make a fuzzy question clear

Write each fuzzy question as a clear one you could check with a ruler or by counting.

"Is it big?" → _____

"Is it long?" → _____

"Does it have lots of spots?" → _____

Sample rewrites

"Is it big?" becomes "Is it wider than 2 cm?"
"Does it have lots of spots?" becomes "Does it have more than 3 spots?"

Puzzle 2.1: Wings

Which bugs have wings? Write their letters. _____

Puzzle 2.2: Legs

Every bug has six legs. Will asking "six legs?" help sort the tray? _____ Why? _____

Puzzle 2.3: Antennae

Which bugs have long antennae? _____

Puzzle 2.4: A poor feature

Why is "Is it pretty?" a poor feature for sorting? _____

Puzzle 2.5: Clearer

Which question is clearer? Circle it.

Is it big? **Is it longer than 2 cm?**

Puzzle 2.6: Spots

How many bugs have spots? _____ Which are they? _____

Puzzle 2.7: Two features

Which bugs have wings *and* spots? _____

Puzzle 2.8: A feature that does not matter

You want to sort the bugs by who can fly, assuming only winged bugs can. Why does spots not help? _____

Puzzle 2.9: Wings or not?

Is "has wings" a clear feature? Explain what would make it unclear. _____

Puzzle 2.10: Your features

List three features of a pile of leaves you could check with a ruler or by counting. _____

Feature tally

Choose ten leaves, shells or small things. Tally each feature and decide whether it is a good splitter.

Feature	Have it (of 10)	Do not have it (of 10)	Good splitter?

Which feature split the ten most evenly? _____

Which feature was hardest to judge? _____

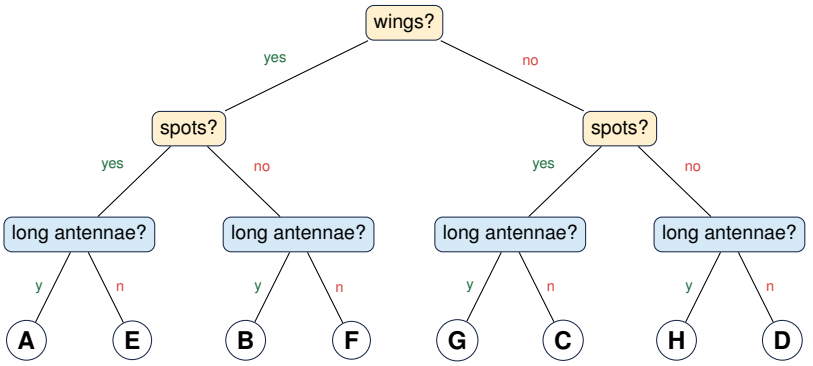
Draw two of your things and mark the feature that tells them apart.

3 A Yes-or-No Key

Splitting a group with one question at a time

A **key** is a list of yes-or-no questions that sorts a thing into a group. You answer the first question. The answer tells you which question to ask next. When there are no more questions, you have an answer.

Every yes-or-no question splits one group into two smaller groups. Here is a key that sorts all eight bugs, one question at a time.



Try it: Follow the key

Pick a bug from the tray on page 10 and follow the key from the top. Write the answers.

Bug _____ Wings? _____ Spots? _____ Long antennae? _____ Ends at: _____

Bug _____ Wings? _____ Spots? _____ Long antennae? _____ Ends at: _____

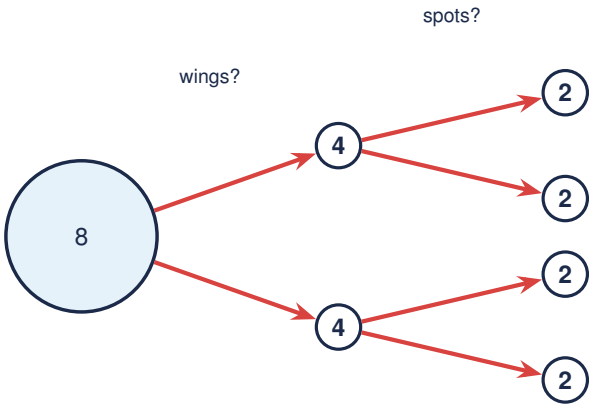
An example

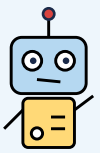
Bug G has no wings, has spots and has long antennae. Wings? no. Spots? yes. Long antennae? yes. It ends at G. Each bug follows its own path, and no two bugs share one.

How many questions?

Questions in a row	Most groups you can make
1	2
2	4
3	8

Each question doubles the number of groups, as long as each question really splits every group.





My key has three questions, and every bug I tried landed in the right place. So every key with three questions must sort every pile correctly. I tried it on eight bugs. Is that enough?

What makes this correction trustworthy?

What we checked. We tried the key on every bug and checked that each ended in a different place.

What this check gives us reason to trust. Eight bugs ended in eight different groups.

What it does not show. That the key works for bugs that are not in the tray. A new bug may have a feature the questions do not cover.

What did we keep?

We turned a pile of bugs into a tree of questions. **We kept** _____ **and we set aside** _____.

Puzzle 3.1: Follow the key

A bug has no wings, has spots and has short antennae. Where does the key end? _____

Puzzle 3.2: How many groups?

Two questions in a row can make at most how many groups? _____
Three questions? _____

Puzzle 3.3: Sixteen bugs

How many questions in a row do you need to sort 16 different bugs, one to a group?
_____

Puzzle 3.4: Fill the gap

A key splits four things A, B, C and D. Question 1: has legs? yes (A, B), no (C, D). Question 2 for the *yes* group must separate A from B. Write a question that would.
_____

Puzzle 3.5: A wasted question

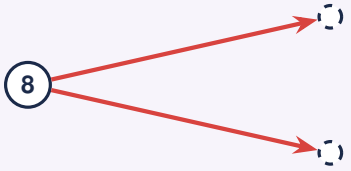
A key has the question "Does it have six legs?" for the bug tray. Why is that a wasted question? _____

Puzzle 3.6: Order of the questions

If the key asks "long antennae?" first, then "spots?", then "wings?", does it still sort all eight bugs? _____

Puzzle 3.7: Draw the circles

A group of 8 is split by spots into two groups. Fill in the empty circles.



Puzzle 3.8: Four things

You have four toys. How many questions might you need in a row? _____

Puzzle 3.9: Make a key

Pick four small things from home. Write a two-question key that sorts them, one to a group. _____

Puzzle 3.10: Checking the end

When your key ends, how many things should be in each last group if you want to tell every one apart? _____

My own key

Draw a key for six or eight things of your own. Put the questions in boxes, the answers on the lines, and the things at the bottom.



Number of questions in a row: _____ Number of groups at the bottom: _____

Does every group have one thing in it? _____

4 Trading Keys

Seeing whether someone else gets the same answer

A key can seem perfect to the person who made it, because the maker knows what every question means. A test of a key is whether *someone else* using it gets the same answer.

Try it: Trade keys with a friend

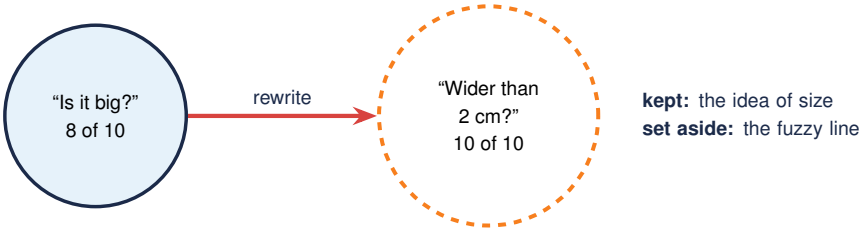
You need: ten buttons or small things, a key you wrote, and a friend.

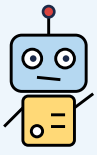
1. Write a key with three yes-or-no questions for your ten things. Sort the ten things yourself and note where each ends.
2. Give your friend the key and the ten things, without telling them where each goes.
3. Ask your friend to sort them using only the key.
4. Count how many things ended in the same place for both of you.

	Things sorted	Same group both times	Different group
Count	10		

Practice numbers, not from a real test

Ten buttons. My friend agreed with me on 8 and disagreed on 2. Agreement: 8 of 10.
The two disagreements were both on the question "Is it big?" My friend and I drew the line in different places.
I rewrote the question as "Is it wider than 2 cm?" and tried again. This time my friend agreed on all 10.





My key sorted my buttons perfectly. I tried it myself, so I know it works. There is no need to ask a friend. Do you agree?

What makes this correction trustworthy?

What we checked. A different person used the key on the same things without being told the answers.

What this check gives us reason to trust. Their groups matched mine on 8 of 10, and after the rewrite on 10 of 10.

What it does not show. That the key works for things neither of us tried, or that two other people would also agree.

What did we keep?

We changed a fuzzy question to a clear one. **We kept** _____ **and we set aside** _____.

Which questions cause trouble?

Question	Clear or fuzzy?	Fix
Is it big?	fuzzy	Is it wider than 2 cm?
Is it dark?	fuzzy	Is it darker than this card?
Does it have holes?	clear	none needed
Is it a nice colour?	fuzzy	drop it, or name colours

What makes this correction trustworthy?

What we checked. We found the questions that made two people disagree and replaced each with one both could check.

What this check gives us reason to trust. After the fix, the same two people got the same answers on all ten things.

What it does not show. That the fix is perfect. A new thing might fall right on the 2 cm line.

Try it: Find the fuzzy question

Look at your own key. Which question did your friend answer differently from you?

_____

How could you change it so a ruler or a count would decide?

_____

Puzzle 4.1: Agreement

A friend agreed on 8 of 10 things. How many disagreements? _____

Puzzle 4.2: After the fix

After the rewrite the friend agreed on 10 of 10. By how many did the agreement improve?
_____

Puzzle 4.3: The unclear question

Which question probably caused the disagreement? Circle it.
Does it have holes? **Is it big?** **Is it round?**

Puzzle 4.4: Spot the unfair test

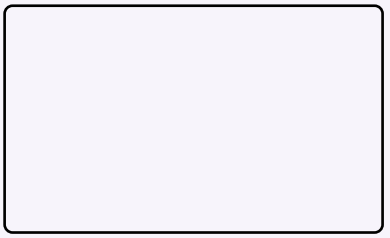
Ana told her friend which group each button belonged in and then asked her to use the key. They agreed on every button. Why does this not show the key is good?
_____

Puzzle 4.5: Three friends

Three friends agreed with the maker on 7, 9 and 8 of ten things. Total _____ Average
_____

Puzzle 4.6: Wrap the question

Draw a circle around “Is it big?” and write the better question outside.



Puzzle 4.7: Two keys

Key X and Key Y both sort the eight bugs into eight groups. Is one of them wrong? _____ How would you decide which is better? _____

Puzzle 4.8: Clear or fuzzy?

Write C for clear or F for fuzzy.
(a) Has four legs _____ (b) Is bright _____ (c) Longer than 5 cm _____ (d) Looks happy _____

Puzzle 4.9: Not told

Why must your friend not be told the answers before using the key? _____

Puzzle 4.10: Your trade

Trade keys with a friend. How many of ten matched? _____ Which question was the problem? _____

Trade record

Record each trade. Write the question that caused disagreement and the change you made.

Trade	Agreed (of 10)	Question that caused trouble	My fix
1			
2			
3			
4			

After the last fix, did a new friend agree on all ten? _____

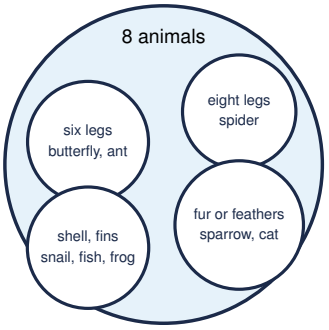
What is one thing you learned about writing questions? _____

5 Groups Inside Groups

Sorting real animals into smaller and smaller circles

Real animals are sorted the same way as bugs, with groups inside groups. All the animals with six legs form a group. Inside that group are smaller groups, such as ones with big colourful wings. Here is a key for eight animals.

No.	Question	Yes, then / No, then
1	Does it have six legs?	go to 2 / go to 3
2	Does it have large, colourful wings?	butterfly / ant
3	Does it have eight legs?	spider / go to 4
4	Does it have feathers or fur?	go to 5 / go to 6
5	Does it have feathers?	sparrow / cat
6	Does it have a hard shell on its back?	snail / go to 7
7	Does it have fins?	goldfish / frog



Each smaller circle sits inside the big one. A good key makes circles that do not overlap.

Try it: Try the key on a walk

You need: paper and a pencil, or a camera, and a safe place to look.

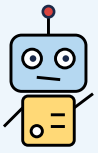
1. Look for three or more animals outside, from a safe distance. Draw or photograph each. Do not touch or pick up any animal.
2. For each, follow the key from question 1. Write where it ends.
3. If an animal does not fit, write down what question it broke.

Animal I saw	Where the key ended	Did it fit?

■ **Adult help:** Look from a safe distance. Do not touch plants or animals you cannot name, and wash hands afterwards.

Practice example

A sparrow: Q1 six legs? no. Q3 eight legs? no. Q4 feathers or fur? yes. Q5 feathers? yes. It ends at sparrow, after 4 questions.
A frog: Q1 no, Q3 no, Q4 no, Q6 no, Q7 no. It ends at frog, after 5 questions.



A dog has fur and no feathers, so the key puts it in the cat group. That means the key calls a dog a cat. I think the key is broken. Is it?

What makes this correction trustworthy?

What we checked. We followed the key for two animals, counted the questions and checked each ended in a group of its own.

What this check gives us reason to trust. The sparrow ended in 4 questions and the frog in 5, and each ended in a different place.

What it does not show. That a dog or a worm would end somewhere sensible. The key only knows the eight animals it was built for.

What did we keep?

We used a key for eight animals. **We kept** _____ **and we set aside** _____.

Puzzle 5.1: An ant

Follow the key for an ant. Write the question numbers you visit. _____

Puzzle 5.2: A frog

How many questions does the frog need? _____

Puzzle 5.3: A goldfish

Write the question numbers for a goldfish. _____

Puzzle 5.4: The fewest

Which animal needs the fewest questions? _____ How many? _____

Puzzle 5.5: Counting a group

How many of the eight animals have six legs? _____ How many do not? _____

Puzzle 5.6: A dog

Where does the key end for a dog? _____ What does that tell you about the "cat" group? _____

Puzzle 5.7: Spiders and insects

Insects have six legs. Use the key to say whether a spider is an insect. _____ Why? _____

Puzzle 5.8: Another way to sort

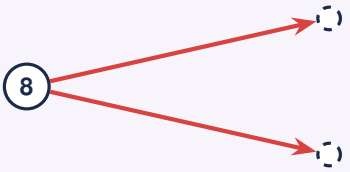
Sort the eight animals by where they live instead: water, land or air. Which feature did you use? _____ Is it the same as the key's groups? _____

Puzzle 5.9: Your walk

List three animals from your walk and where the key put each. _____

Puzzle 5.10: Draw the circles

Eight animals are split by six legs into 2 and 6. Fill in the circles.



Animal log

Record the animals you see, and follow the key for each.

Animal	Where I saw it	Where the key ended	Did it fit?

Which animal broke the key, if any? _____

Draw one animal you saw.

6 When Something Does Not Fit

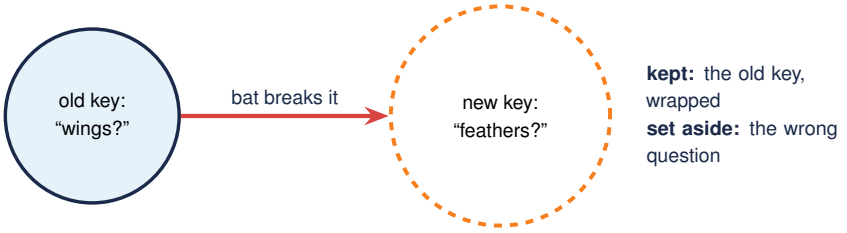
Changing the key, not forcing the fit

Every key is built from the things its maker had in front of them. Sooner or later a thing turns up that does not fit. There are two things you can do. You can force it into the nearest group. Or you can change the key.

Model	What it keeps	What it drops
A feature	one thing about each item	every other thing
A yes-or-no key	a path of answers	how items differ in other ways
A group	what the members share	how they differ
A nested circle	groups inside groups	the details of each item

The bat that broke a key

Suppose a key says: "Does it have wings? Yes, it is a bird. No, it is a mammal." A bat has wings, so the key calls it a bird. But a bat has fur and feeds its young milk. The key was built for the birds and the cats it knew.



Try it: Break your own key

Take the key you wrote in Chapter 4 or 5. Find or imagine one thing that does not fit it. What happens? Write the new question that fixes it.

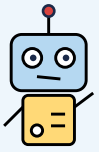
The thing that broke it: _____

Which question failed? _____

The new question: _____

The worm in the animal key

In Chapter 5, a worm has no legs, no feathers or fur, no shell and no fins. Q1 no, Q3 no, Q4 no, Q6 no, Q7 no. It ends at frog. But a worm is not a frog.
The fix: add a new question before Q7, "Does it have legs?" A worm answers no and goes to a new group. The old key is wrapped by the new one.



The worm ended in the frog group. The key must be right, so the worm must be a frog. I trust the key. Do you?

What makes this correction trustworthy?

What we checked. We tried a new specimen on the old key, saw where it ended and fixed the key when the result looked wrong.

What this check gives us reason to trust. The worm and the bat each exposed a missing question, and the fixes kept every earlier animal where it was.

What it does not show. That the new key is finished. Some other animal will break it too. A key is always a model for the things it has met.

What did we keep?

We changed a key when something did not fit. **We kept** _____ **and we set aside** _____.

Puzzle 6.1: The bat

A key says "wings? yes: bird, no: mammal". Where does it put a bat? _____
Why is that wrong? _____

Puzzle 6.2: The worm

Which new question separates the worm from the frog?
_____

Puzzle 6.3: Change or force?

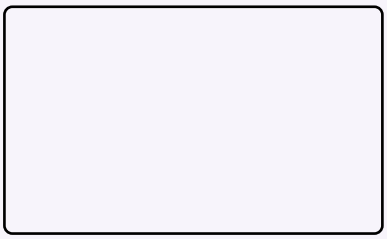
Is it better to force a surprise thing into the nearest group or to change the key? _____ Why? _____

Puzzle 6.4: A new bug

A new bug has wings, spots and antennae that are neither long nor short. Does the bug key sort it? _____ What would you change? _____

Puzzle 6.5: Wrap the key

Draw the new question wrapped around the old key.



Puzzle 6.6: Eight groups

A key uses wings, spots and antennae. How many different groups can it make? _____

Puzzle 6.7: Already there

A new bug J has no wings, no spots and long antennae. Which bug in the tray matches J exactly? _____

Puzzle 6.8: Sentence check

Mark S if the evidence supports it, G if it is a guess, and B if it is too big.
(a) “The key put the bat with the birds.” _____
(b) “Wings are a poor feature for sorting animals.” _____
(c) “No key works for every animal ever.” _____

Puzzle 6.9: Flying

Birds and bats both fly. What feature would separate them? _____

Puzzle 6.10: Thirty-two things

You need to separate 32 things, one to a group. How many questions in a row? _____

Puzzle 6.11: Model or real?

Write M for model and R for real thing.
(a) the key _____ (b) the bat _____ (c) the tray picture _____

Puzzle 6.12: Agreement

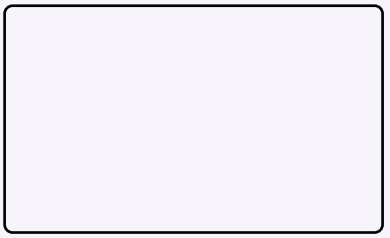
Two people used a key on ten things and landed in the same group on all ten. Does that prove the key is perfect? _____ Why not? _____

Puzzle 6.13: Trust

A student says her key is perfect because she used it herself. What would you ask her to do? _____

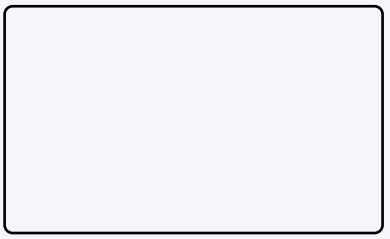
Puzzle 6.14: Draw a key for four

Draw a two-question key for four things you choose. Label the questions.



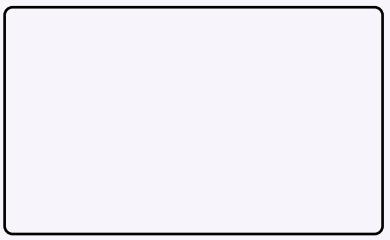
Puzzle 6.15: Draw the split

Draw a circle of 16 splitting into two of 8, and one of the 8s splitting again.



Puzzle 6.16: Draw the wrap

Draw a circle around “wings? bird” and write the fixed question outside.

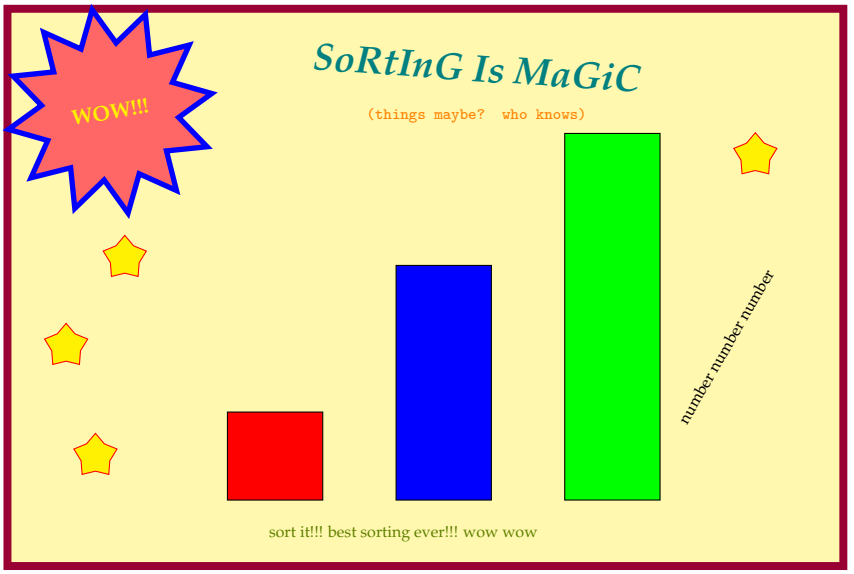


Slop check

Slop is hasty, gaudy work that forgot to take care. It can be made by a person or a machine, quickly or slowly. What matters is whether the work shows care for three things:

- **Salient:** Is the main point easy to see?
- **Simple:** Is there anything that does nothing but get in the way?
- **Relevant:** Does everything on the page help answer the question?

Here is a hasty poster about sorting. Study it, then decide what is wrong.



Slop check

(a) What is the poster's main point? Can you tell?

_____

(b) Name two things that are not simple, because they only distract.

_____

(c) Name two things that are missing, so the bars cannot be read.

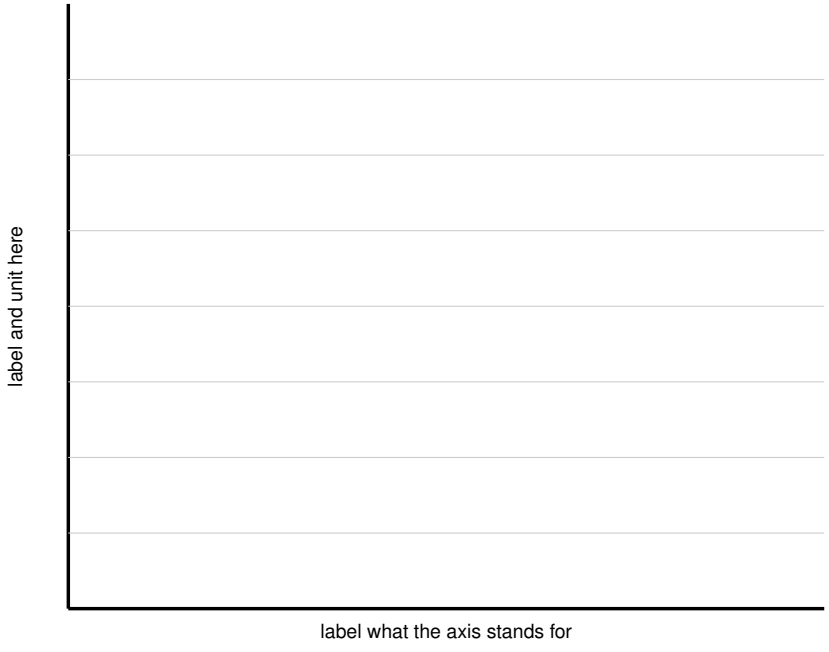
_____

(d) Which part is not relevant to sorting at all?

_____

Now you try

On a separate sheet, redraw a clear bar chart of how the twelve buttons were sorted by colour: red 3, blue 4 and white 5. Give it a title, label the axes with units, use one or two calm colours, and delete everything that does not help. Check it against the three words: salient, simple, relevant.

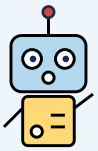


Title:

What did we keep, across the whole book?

For each chapter, say one thing that was simplified, what was kept, and what was set aside. A sample is done for you.

Chapter	What we simplified	What we kept	What we set aside
1 Many ways to sort	a pile as groups	the feature we chose	every other feature
2 Features that matter	a bug as a few features	clear splitters	fuzzy features
3 A yes-or-no key	a sort as questions	the path of answers	other differences
4 Trading keys	a key as shared rules	agreement between people	fuzzy lines
5 Groups inside groups	animals as nested circles	the groups that nest	details of each animal
6 When something does not fit			



Everything you sorted in this book was a simplification of something. A key is allowed to drop things, as long as you say which. Next time something does not fit, you will change the key.

Write a sorting report

1. **What I looked at.**
2. **What I changed, measured and kept the same.**
3. **My readings.** All of them.
4. **My answer.** With how many agreed.
5. **What I kept and what it hides.**

Pick one key from this book and write a report on it. Use the lines below.

Tests to try

Pick one and design it with the habits from this book: change one thing, keep the rest the same, repeat, and check with a friend.

1. Does a key with fewer questions agree as often as one with more?
2. Do two friends sort the same ten leaves the same way?
3. Which feature best splits a pile of shells: colour, shape or size?
4. Can a younger child use your key and agree with you?
5. How many questions does a key need for 20 things?
6. Does the order of the questions change how many things end up in each group?

Write the test you pick:

_____

_____

_____

_____

_____

_____

Change: _____ Measure: _____ Keep the same: _____

A week of sorting

Each day, notice one group of things and the feature that makes it a group.

Day	The group	The feature
Mon		
Tue		
Wed		
Thu		
Fri		
Sat		
Sun		

Questions I still have

Write what you still wonder about sorting. Then pick one and think how to test it.

Answers

Try every puzzle first. Where an answer shows a calculation, check the steps, not just the result.

Chapter 1

Multix on the one way. No. Each feature gives different groups and none is the right one. Which is best depends on what you want to find out. **Keep box.** Kept: the colour of each button. Set aside: size, shape and holes.

1.1 $3 + 4 + 5 = 12$. Yes.
1.2 $7 + 4 = 11$. One button was left out or not counted.
1.3 The sort by colour made more groups, 3 against 2.
1.4 Examples: size, colour, laces, material.
1.5 Different features make different groups, and which sort is best depends on the question.
1.6 The empty circle is 12.
1.7 4 have wings (A, B, E, F) and 4 do not.
1.8 Yes. For example, the only gold button in the pile.
1.9 In one sort a button is in one group only. It can be in "red" in a colour sort and "four holes" in a holes sort.
1.10 Answers vary. Check the counts add up.

Chapter 2

Multix on six legs. No. A feature every bug has splits nothing. **Keep box.** Kept: whether a bug has wings. Set aside: spots and antennae.

2.1 A, B, E, F.
2.2 No. All eight bugs answer yes, so they stay in one group.
2.3 A, B, G, H.
2.4 It is an opinion, and people disagree.
2.5 "Is it longer than 2 cm?"
2.6 4: A, C, E, G.
2.7 A and E.
2.8 Spots do not change whether a bug has wings.
2.9 Not always. A tiny or folded wing may be unclear, so say what counts as a wing.
2.10 Answers vary.

Chapter 3

Multix on three questions. No. Eight bugs may not include one with an unexpected feature. **Keep box.** Kept: the path of answers. Set aside: other differences.

3.1 C.
3.2 4. 8.
3.3 4.
3.4 Example: "Does it have a tail?"
3.5 Every bug answers yes, so it does not split the group.
3.6 Yes. Every combination is still covered.
3.7 4 and 4.
3.8 2.
3.9 Answers vary.
3.10 One.

Chapter 4

Multix on trading. No. The maker knows what each question means. Someone else must try it. **Keep box.** Kept: the idea of size. Set aside: the fuzzy line.

4.1 2.
4.2 2 more.

4.3 "Is it big?"
4.4 She told her friend the answers, so the friend was not using the key on her own.
4.5 $7 + 9 + 8 = 24$ and $24 \div 3 = 8$.
4.6 Answers vary. Check the old question is circled and a clear one is outside.
4.7 Neither is necessarily wrong. Compare how clear the questions are and how many people agree.
4.8 (a) C (b) F (c) C (d) F.
4.9 If they know the answers, their sorting may only copy them, which does not test the key.
4.10 Answers vary.

Chapter 5

Multix on the dog. The key is not broken. The "cat" end really means fur without feathers, and the key only knew a cat. **Keep box.** Kept: groups that nest. Set aside: details of each animal.

5.1 1 and 2.
5.2 5.
5.3 1, 3, 4, 6 and 7.
5.4 Butterfly and ant, 2 questions each.
5.5 2 have six legs and 6 do not.
5.6 It ends at cat. The "cat" group is really fur without feathers.
5.7 No. A spider has eight legs and the key puts it in the eight-legs group.
5.8 Answers vary, for example goldfish in water, sparrow in air, snail on land. These groups do not match the key's.
5.9 Answers vary.
5.10 2 and 6.

Chapter 6

Multix on the worm. No. The key had no question for a worm, so it was forced into the frog group. Change the key. **Keep box.** Kept: the old key, wrapped. Set aside: the wrong question.

6.1 Bird. A bat has fur and feeds its young milk.
6.2 "Does it have legs?"
6.3 Change the key. Forcing the fit hides the problem.
6.4 No. "Neither long nor short" is unclear. Add a rule such as "longer than 1 cm".
6.5 Answers vary. Check the old key is wrapped by the new question.
6.6 8.
6.7 H.
6.8 (a) S (b) G (c) B.
6.9 Feathers.
6.10 5, since $2^5 = 32$.
6.11 (a) M (b) R (c) M.
6.12 No. Only ten things were tried, and new things may break it.
6.13 Ask her to trade keys with a friend and compare.
6.14 to 6.16 Answers vary. Check each split matches its labels.
Slop check. (a) You cannot tell: no title, no labels. (b) Many colours, stars, joke text. (c) A title, the colours and the counts with units. (d) The stars and jokes.

Words

feature one thing you can check about an item, such as wings or colour.

group things that share a feature.

key a list of yes-or-no questions that sorts something into a group.

model a simple stand-in that keeps some things and drops others.

nested one group inside another.

specimen one example of a thing you are sorting.

Notes for grown-ups

This book teaches that a classification is a model: it keeps the features chosen for sorting and drops the rest. A key is tested by whether a second person gets the same answers, and a key is changed, not forced, when a surprise specimen does not fit. Each chapter has one activity, one worked example with made-up practice numbers, one trust box, one "what did we keep?" prompt and a page of puzzles.

Materials and safety. Buttons or small objects, leaves or shells, paper, a ruler, a friend and a safe place to look at animals. Do not touch or pick up animals, and do not touch plants you cannot name. Keep small objects away from young children and pets.

The eight bugs are made up so that every combination of three features appears once. They are not real insects. The animal key in Chapter 5 is a teaching key, and real classification uses many more features.

The practice numbers were made up so the arithmetic is easy to check.

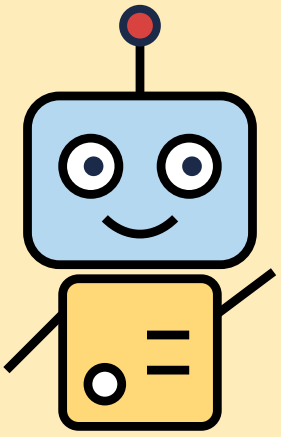
Sources

- N. A. Campbell, J. B. Reece and others. *Campbell Biology*. Pearson. For classification and the grouping of living things.
- J. R. Taylor. *An Introduction to Error Analysis*, 2nd edition. University Science Books, 1997. For agreement between independent readings.

Next in Science

Where the Energy Goes

A ball rolls down a ramp, across the floor, and stops. Book 8 follows the ball's energy from start to finish and keeps a running total of where it went.



Ages 9 to 11 ⋆ Science Book 7